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Aufgabe 1.1

Let $N = (S, T, W)$ be a Petri net with connectivity matrix $\mathbb{C}$ and $M_1, M_2 \in \mathbb{N}^{|S|}, \sigma = t_i \dots t_m \in T^*$ such that $M_1[\sigma\rangle M_2$. To prove: $M_2 = M_1 + \mathbb{C} \cdot \psi(\sigma)$

Proof. From the definition of the connectivity matrix it follows that:

$$\mathbb{C} = \Big(W(t_1, -) - W(-, t_1) \dots W(t_{|T|}, -) - W(-, t_{|T|})\Big)$$

Thus the term $\mathbb{C} \cdot \psi(\sigma)$ results in

$$\mathbb{C} \cdot \psi(\sigma) = (W(t_i, -) - W(-, t_i) + \dots + W(t_m, -) - W(-, t_m))$$

and therefore according to the definition of the firing relation:

$$M_1[\sigma\rangle M_2 = M_1 + W(t_i, -) - W(-, t_i) + \dots + W(t_m, -) - W(-, t_m) = M_1 + \mathbb{C} \cdot \psi(\sigma)$$

Thus the proposition is proven. □

Aufgabe 1.2

Aufgabe 1.3

(a)

Let (N, M_0, M_f) be a petri-net with $N = (S, T, W)$.

Proof. □

(b)

Let $A = (Q, \Delta, q_0, v_0)$ be a VASS of dimension $d \in \mathbb{N}$. We construct an equivalent petri-net as follows:
$N = (S, T, W)$ with

- $S = \{s_1, \dots s_d, s_q\}$
- $T = \{t_\delta \mid \forall \delta \in \Delta\}$

The weight function $W : (S \times T) \cup (T \times S) \to \mathbb{N}$ is defined for any $\delta = (q_k, a, q_l) \in \Delta$:

$$W(s_i, t_\delta) = \begin{cases} -a_i & \text{if } a_i < 0 \\ 0 & \text{otherwise} \end{cases}, \quad W(t_\delta, s_i) = \begin{cases} a_i & \text{if } a_i > 0 \\ 0 & \text{otherwise} \end{cases}$$

and

$$W(t_\delta, s_q) = \begin{cases} (l - k) & \text{if } l > k \\ 0 & \text{otherwise} \end{cases}, \quad W(s_q, t_\delta) = \begin{cases} (k - l) & \text{if } k > l \\ 0 & \text{otherwise} \end{cases}$$

The initial marking is $M_0 = \begin{pmatrix} v_{0_1} & \dots & v_{0_d} & 0 \end{pmatrix}^T$ and the final marking $M_f = \begin{pmatrix} v_{f_1} & \dots & v_{f_d} & m \end{pmatrix}^T$ where $q_m = q_f$. This petri net is equivalent to the VASS.

Proof. Let $\delta = (q_i, a, q_j)$ be a transition in the VASS with v being the counter assignment before the transition and $(v + a)$ the assignment after the transition. In the formed petri net, the transition t_δ should have the same effect as δ in the VASS, meaning, the marking before the transition should be $M_i = \begin{pmatrix} v_1 & \dots & v_d & i \end{pmatrix}^T$ and after the transition it should be $M_j = \begin{pmatrix} v_1 + a_1 & \dots v_d + a_d & j \end{pmatrix}^T$.

First we prove, that $M_j(s_q) = j$ if $M_i(s_q) = i$. For this we need to inspect two cases:

1. If $i < j$, then $j - i > 0$. Therefore there exists a weight $W(t_\delta, s_q) = j - i$. Therefore, if t_δ is triggered, it adds $(j - i)$ tokens to s_q. Since $i + (j - i) = j$, s_q has j tokens in marking M_j.

2. If $i > j$, then $j - i < 0$. Therefore there exists a weight $W(t_\delta, s_q) = i - j$. Therefore, if t_δ is triggered, it takes $(i - j)$ tokens from s_q. Since $i - (i - j) = i - i + j = j$, s_q has j tokens in marking M_j.

Now, we show the equivalence of the transitions at all other places.

$$
\begin{aligned}
& \delta = (q_i, a, q_j) \in \Delta \wedge v \in \mathbb{N}^d \text{ assignment before transition} \\
\Leftrightarrow \quad & t_\delta \in T \wedge \forall a_i > 0 : W(t_\delta, s_i) = a_i \wedge \forall a_i < 0 : W(s_i, t_\delta) = -a_i \\
\Leftrightarrow \quad & M_j(s_1, \dots, s_d) = M_i(s_1, \dots, s_d) + W(t_\delta, s_1, \dots, s_d) - W(s_1, \dots, s_d, t_\delta) \\
\Leftrightarrow \quad & \forall k \in \{1, ..., d\} : M_j(s_k) = M_i(s_k) + W(t_\delta, s_k) - W(s_k, t_\delta) = v_k + a_k \\
\Leftrightarrow \quad & M_j = \begin{pmatrix} v_1 + a_1 & \dots & v_d + a_d & j \end{pmatrix}
\end{aligned}
$$

Since this shows that any transition δ can be replicated by t_δ in the petri net, if there exists a sequence of transitions $\delta_1, \dots, \delta_m$ with $(q_0, v_0) \xrightarrow{\delta_1} \dots \xrightarrow{\delta_m} (q_f, v_f)$, there has to be a sequence $\sigma = t_{\delta_1} \dots t_{\delta_m}$ with $M_0[\sigma\rangle M_f$ and the other way around. □

(c)

Aufgabe 1.4